

Max Mobley

memobley@wpi.edu | <https://maxmob.sh> | +1 703-635-0410

EXPERIENCE

Formula SAE Electric - Goat Fast Racing

Worcester, MA

Team Lead, Driverless Lead, Driver (Previously: Telemetry Lead)

Aug. 2024 - Present

- Built radio telemetry system, including PCB design (KiCad), PCB assembly, and embedded programming with STM32
- Created realtime telemetry pipeline to parse and visualize 5,000+ events per second with near-zero end-to-end latency
- Performed failure mode analysis for safety critical software and hardware, including pedal sensors and steering wheel
- Worked across teams to minimize roadblocks and ensure integration of hundreds of electrical & mechanical components

Innovative Wireless Technologies

Lynchburg, VA

Software Engineering Intern

May 2025 - Aug. 2025

- Built and iterated on end-of-production test fixtures and comprehensive testing software for safety-critical hardware
- Reduced net test time of existing fixtures by over 70% through software optimization and mechanical improvements
- Designed handheld serial debugger to increase system status visibility for development, testing, and field repair

Worcester Polytechnic Institute

Worcester, MA

Server Hardware Technician - Student Worker

Sep. 2022 - Present

- Serviced and upgraded WPI academic research cluster, including Dell EMC, Arista, and Juniper hardware systems
- Built software tools to assist students in using cluster resources, such as an SSO-enabled RDP authentication proxy

CollabLab Makerspace - Lab Monitor & President

Sep. 2022 - Present

- Assisted students with personal projects and taught them to use tools, CAD, 3D printers, programming, and electronics

NerdSpeak LLC

Remote

Software Engineer

June 2021 - May 2022

- Built software systems which helped over 100,000 users purchase hard-to-find tech items during electronics shortages
- Designed and optimized a high-throughput, distributed software architecture to achieve lowest-latency web scraping

PROJECTS

Autonomous Battlebot: Built robot & supporting software for an AI fighting robot with numerous sensors, PCBs, custom ML models, which altogether powered a ground-up control pipeline with low latency, high accuracy, and complex awareness

Digital Website Archival: Built high throughput web scrapers & reverse engineered web APIs to save decaying websites, often volunteering & working with the Internet Archive to ensure multiple copies are saved and made available to the public

Home Server Infrastructure: Created and maintained fully redundant networking and compute in a home environment and architected multi-hundred-terabyte storage systems to serve as a data warehouse for website archival operations

SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, Bash, Golang

Frameworks & Technologies: S3, SQLite, Ceph, ElasticSearch, git, Unix, Linux, Proxmox, Cisco, Jira, Grafana, VSCode, Docker (Swarm), Solidworks, Flask, KiCAD, STM32, Postgres, NGINX, OpenCV, mitmproxy, Selenium, I2C, ESP-IDF, CAN

Concepts: Networking, Distributed Systems, Backend & REST API Development, CI/CD, Reverse Engineering, High Throughput Web Scraping, PCBA, Database Programming, Computer Vision, Oscilloscope, Logic Analyzer

Soft Skills: Remote & In-Person Experience, Leadership, Collaboration, Self-Teaching, Public Speaking, Technical Writing

EDUCATION

Worcester Polytechnic Institute

Class of 2026

B.S. Computer Science & B.S. Electrical & Computer Engineering | GPA: 3.5

Worcester, MA

Coursework: Operating Systems, Machine Assembly, Algorithms, Database Systems, Computer Networking, Software Engineering, Intro to Computer Engineering, Electrical Systems, Sensors & Circuits, Intro to Robotics Engineering

Research (Lyon, FR - 2024): Conducted interviews, research, and wrote a 130-page report on Lyon's public transit system